

Supplemental Information 2

Description

An R markdown of the phenotypic analysis and relevant information pertaining to interpretation of results. Analysis will be performed and results will be explained after relevant chunk information.

```
# Load necessary libraries
suppressMessages(library(gaston))
suppressMessages(library(tidyverse))
suppressMessages(library(asreml))

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suppressMessages(library(ASMap))
suppressMessages(library(qtl))
suppressMessages(library(psych))
suppressMessages(library(knitr))
suppressMessages(library(ggplot2))
suppressMessages(library(patchwork))

# Read in data
pheno <- read.csv("NC13955_Combined_Data_Final.csv")

# Read in potential selfs
selfs <- read.csv("list_of_potential_selfs.csv")

# Remove selfs
pheno <- pheno %>%
  filter(!GENOTYPE %in% selfs$GENOTYPE)

# Ensure the correct data types
pheno[, 1:7] <- lapply(pheno[, 1:7], as.factor)
pheno[, 8:ncol(pheno)] <- lapply(pheno[, 8:ncol(pheno)], as.numeric)

# Colnames
colnames(pheno)[8:ncol(pheno)] <- c("Heading Date (Days)",
  "Visual Rating (0-9)",
  "Fusarium Damaged Kernels (%)",
  "DON Content (PPM)",
  "Plant Height (cm)")
```

Single environment analysis

Here I am going through each environment and assessing the significance of the genotype effect using the following model

$$y_{ij} = \mu + G_i + r_j + \varepsilon_{ij}$$

Where G is the fixed genotype effect, r is the random replication effect, and ε is the residual error. Each loop will be denoted using “#” as the dilimeter and the environment assessed will be denoted by the assigned short hand (KIN = Kinston, NC; RAL = Raleigh, NC; WAR = Warsaw, VA; 19 = 2019; 20 = 2020). We will use a fully random model to estimate heritability via the following equation:

$$h^2 = \frac{\sigma_g^2}{\sigma_g^2 + \sigma_\varepsilon^2}$$

```
# Set up lists for loop
env <- levels(pheno$ENV)
traits <- colnames(pheno)[8:12]

# Create dataframe for means output
BLUEs_loc <- data.frame(GENOTYPE = unique(pheno$GENOTYPE))
h2_envs <- c()

# Run models for each environment
for (i in env) {
  for (j in traits) {
    # Print message
    print_message <- paste("### Analyzing", j, "in", i, "###")
    print(paste(rep("#", nchar(print_message)), collapse = ""))
    print(print_message)
    print(paste(rep("#", nchar(print_message)), collapse = ""))

    # Read subset of phenotype data
    pheno_subset <- na.omit(pheno[pheno$ENV == i, c("GENOTYPE", "REP", j)])
    colnames(pheno_subset)[3] <- "Y"

    # Check if subset is empty
    if (nrow(na.omit(pheno_subset)) == 0) {
      # Throw message
      print(paste("###", j, "not taken in", i, "moving on..."))
      remove(pheno_subset)
      next
    }

    # If the trait is abnormally distributed
    if (j %in% c("Visual Rating (0-9)",
                "Fusarium Damaged Kernels (%)",
                "DON Content (PPM)")) {
      # Run model and pull estimates
      fit1 <- suppressMessages(
        suppressWarnings(
          asreml(fixed = Y ~ GENOTYPE,
                random = ~ REP,
                residual = ~ units,
                data = pheno_subset,
                family = asr_poisson(),
                maxit = 75,
                trace = FALSE)
        )
      )
    }
  }
}
```

```

      # Run model and pull estimates
fit2 <- suppressMessages(
  suppressWarnings(
    asreml(fixed = Y ~ 1,
          random = ~ GENOTYPE+REP,
          residual = ~ units,
          data = pheno_subset,
          family = asr_poisson(),
          maxit = 75,
          trace = FALSE)
  )
)

# Check if model converges and if it results in critical error,
# default to RCBD with normal distribution
if (fit1$converge == FALSE | (i == "WAR20" &
                             j == "Visual Rating (0-9)")) {

  # Fit model
  fit1 <- suppressMessages(
    suppressWarnings(
      asreml(fixed = Y ~ GENOTYPE,
            random = ~ REP,
            residual = ~ units,
            data = pheno_subset,
            maxit = 75,
            trace = FALSE)
    )
  )

  # Fit model
  fit2 <- suppressMessages(
    suppressWarnings(
      asreml(fixed = Y ~ 1,
            random = ~ GENOTYPE+REP,
            residual = ~ units,
            data = pheno_subset,
            maxit = 75,
            trace = FALSE)
    )
  )

  # Update model
  fit1 <- update.asreml(fit1)
  fit2 <- update.asreml(fit2)
}

# Print message
print_message <- paste("### Summary of fixed effect model for",
                       j,
                       "in",
                       i,
                       "###")
print(paste(rep("#", nchar(print_message)), collapse = ""))
print(print_message)

```

```

print(paste(rep("#", nchar(print_message)), collapse = ""))

# Get quick summary (comment out for markdown)
print(summary(fit1))

# Pull BLUEs
pred <- predict(fit1, classify = "GENOTYPE", maxit = 75)$pvals[, c("GENOTYPE",
                                                                    "predicted.value")]

# Rename
colnames(pred)[2] <- paste(j, "_", i, sep = "")

# Bind
BLUES_loc <- left_join(BLUES_loc, pred, by = "GENOTYPE")

# Print message
print_message <- paste("### Summary of random effect model for",
                        j,
                        "in",
                        i,
                        "###")
print(paste(rep("#", nchar(print_message)), collapse = ""))
print(print_message)
print(paste(rep("#", nchar(print_message)), collapse = ""))

# Print summary
print(summary(fit2))

# Estimate heritability
h2 <- vpredict(fit2, h2~V2/(V2+V3))

# Print message
print_message <- paste("### Narrow-sense heritability of",
                        j,
                        "in",
                        i,
                        "###")
print(paste(rep("#", nchar(print_message)), collapse = ""))
print(print_message)
print(paste(rep("#", nchar(print_message)), collapse = ""))

# Table
print(knitr::kable(h2, digits = 2))

# Add to h2
h2_envs <- rbind(h2_envs,
                 data.frame(Environment = i,
                             Trait = j,
                             h2))
} else {
  # Run model and pull estimates
  fit1 <- suppressMessages(

```

```

suppressWarnings(
  asreml(fixed = Y ~ GENOTYPE,
    random = ~ REP,
    residual = ~ units,
    data = pheno_subset,
    trace = FALSE)
)
)

# Run model and pull estimates
fit2 <- suppressMessages(
  suppressWarnings(
    asreml(fixed = Y ~ 1,
      random = ~ GENOTYPE+REP,
      residual = ~ units,
      data = pheno_subset,
      trace = FALSE)
  )
)

# Print message
print_message <- paste("### Summary of fixed effect model for",
  j,
  "in",
  i,
  "###")
print(paste(rep("#", nchar(print_message)), collapse = ""))
print(print_message)
print(paste(rep("#", nchar(print_message)), collapse = ""))

# Get quick summary (comment out for markdown)
print(summary(fit1))

# Get BLUES
pred <- predict(fit1, classify = "GENOTYPE")$pvals[, c("GENOTYPE",
  "predicted.value")]

# Rename
colnames(pred)[2] <- paste(j, "_", i, sep = "")

# Bind
BLUES_loc <- left_join(BLUES_loc, pred, by = "GENOTYPE")

# Print message
print_message <- paste("### Summary of random effect model for",
  j,
  "in",
  i,
  "###")
print(paste(rep("#", nchar(print_message)), collapse = ""))
print(print_message)
print(paste(rep("#", nchar(print_message)), collapse = ""))

```

```

# Print summary
print(summary(fit2))

# Estimate heritability
h2 <- vpredict(fit2, h2~V2/(V2+V3))

# Print message
print_message <- paste("### Narrow-sense heritability of",
                        j,
                        "in",
                        i,
                        "###")
print(paste(rep("#", nchar(print_message)), collapse = ""))
print(print_message)
print(paste(rep("#", nchar(print_message)), collapse = ""))

# Table
print(knitr::kable(h2, digits = 2))

# Add to h2
h2_envs <- rbind(h2_envs,
                 data.frame(Environment = i,
                             Trait = j,
                             h2))
}
# Remove temporary objects
remove(pheno_subset, fit1, fit2, pred, h2)
}
}

```

```

## [1] "#####"
## [1] "### Analyzing Heading Date (Days) in KIN19 ###"
## [1] "#####"
## [1] "### Heading Date (Days) not taken in KIN19 moving on..."
## [1] "#####"
## [1] "### Analyzing Visual Rating (0-9) in KIN19 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 23.72
##          : Variance heterogeneity factor (Deviance/df): 0.18 assuming 131 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 131.72
##          : Variance heterogeneity factor (Deviance/df): 0.45 assuming 295 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Visual Rating (0-9) in KIN19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##

```

```

## $loglik
## [1] 23.57785
##
## $nedf
## [1] 131
##
## $sigma
## [1] 1
##
## $varcomp
##           component std.error z.ratio bound %ch
## REP      1.011929e-07      NA      NA      B  NA
## units!R  1.000000e+00      NA      NA      F   0
##
## $bic
## [1] -42.28049
## attr("parameters")
## [1] 1
##
## $aic
## [1] -45.15569
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 23.72133
##
## $heterogeneity
## [1] 0.1810788
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 23.72
##           : Variance heterogeneity factor (Deviance/df): 0.18 assuming 131 df.
## [1] "#####"
## [1] "### Summary of random effect model for Visual Rating (0-9) in KIN19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       family = asr_poisson(), data = pheno_subset, maxit = 75,
##       trace = FALSE)
##
## $loglik
## [1] 91.22795
##
## $nedf
## [1] 295
##
## $sigma
## [1] 1
##
## $varcomp

```

```

##          component  std.error  z.ratio bound %ch
## REP      1.011929e-07      NA      NA      B  NA
## GENOTYPE 3.236618e-02 0.01749005 1.850548      P 0.3
## units!R  1.000000e+00      NA      NA      F 0.0
##
## $bic
## [1] -171.0819
## attr(,"parameters")
## [1] 2
##
## $aic
## [1] -178.4559
## attr(,"parameters")
## [1] 2
##
## $deviance
## [1] 131.7225
##
## $heterogeneity
## [1] 0.446517
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Visual Rating (0-9) in KIN19 ###"
## [1] "#####"
##
##
## | | Estimate| SE|
## |--|-----:|----:|
## |h2 |      0.03| 0.02|
## [1] "#####"
## [1] "### Analyzing Fusarium Damaged Kernels (%) in KIN19 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 823.20
##          : Variance heterogeneity factor (Deviance/df): 5.92 assuming 139 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 845.84
##          : Variance heterogeneity factor (Deviance/df): 2.79 assuming 303 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Fusarium Damaged Kernels (%) in KIN19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##
## $loglik
## [1] -199.3016
##
## $nedf

```

```

## [1] 139
##
## $sigma
## [1] 1
##
## $varcomp
##      component      std.error    z.ratio bound %ch
## REP      0.001413535 0.002233495 0.6328804    P 2.5
## units!R 1.000000000          NA          NA    F 0.0
##
## $bic
## [1] 403.5377
## attr(,"parameters")
## [1] 1
##
## $aic
## [1] 400.6032
## attr(,"parameters")
## [1] 1
##
## $deviance
## [1] 823.1962
##
## $heterogeneity
## [1] 5.922275
##
## attr(,"class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 823.19
##          : Variance heterogeneity factor (Deviance/df): 5.92 assuming 139 df.
## [1] "#####"
## [1] "### Summary of random effect model for Fusarium Damaged Kernels (%) in KIN19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##
## $loglik
## [1] -179.0807
##
## $nedf
## [1] 303
##
## $sigma
## [1] 1
##
## $varcomp
##      component      std.error    z.ratio bound %ch
## REP      0.001510959 0.002443377 0.6183898    P 0
## GENOTYPE 0.263990173 0.032279805 8.1781835    P 0
## units!R 1.000000000          NA          NA    F 0

```

```

##
## $bic
## [1] 369.5888
## attr("parameters")
## [1] 2
##
## $aic
## [1] 362.1614
## attr("parameters")
## [1] 2
##
## $deviance
## [1] 845.8436
##
## $heterogeneity
## [1] 2.791563
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Fusarium Damaged Kernels (%) in KIN19 ###"
## [1] "#####"
##
##
## | | Estimate| SE|
## |--|-----:|----:|
## |h2 | 0.21| 0.02|
## [1] "#####"
## [1] "### Analyzing DON Content (PPM) in KIN19 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 571.66
## : Variance heterogeneity factor (Deviance/df): 4.11 assuming 139 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 595.24
## : Variance heterogeneity factor (Deviance/df): 1.96 assuming 303 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for DON Content (PPM) in KIN19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
## family = asr_poisson(), data = pheno_subset, maxit = 75,
## trace = FALSE)
##
## $loglik
## [1] -130.6611
##
## $nedf
## [1] 139
##
## $sigma
## [1] 1

```

```

##
## $varcomp
##      component  std.error  z.ratio bound %ch
## REP      0.009194348 0.01132679 0.811735    P 11.4
## units!R 1.000000000      NA      NA      F 0.0
##
## $bic
## [1] 266.2566
## attr(,"parameters")
## [1] 1
##
## $aic
## [1] 263.3221
## attr(,"parameters")
## [1] 1
##
## $deviance
## [1] 571.6594
##
## $heterogeneity
## [1] 4.112658
##
## attr(,"class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 571.65
##          : Variance heterogeneity factor (Deviance/df): 4.11 assuming 139 df.
## [1] "#####"
## [1] "### Summary of random effect model for DON Content (PPM) in KIN19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##
## $loglik
## [1] -157.4584
##
## $nedf
## [1] 303
##
## $sigma
## [1] 1
##
## $varcomp
##      component  std.error  z.ratio bound %ch
## REP      0.008943393 0.01320625 0.677209    P 0
## GENOTYPE 0.443225934 0.05508508 8.046207    P 0
## units!R 1.000000000      NA      NA      F 0
##
## $bic
## [1] 326.3443
## attr(,"parameters")

```

```

## [1] 2
##
## $aic
## [1] 318.9168
## attr(,"parameters")
## [1] 2
##
## $deviance
## [1] 595.2395
##
## $heterogeneity
## [1] 1.964487
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of DON Content (PPM) in KIN19 ###"
## [1] "#####"
##
##
## | | Estimate| SE|
## |--|-----:|----:|
## |h2 | 0.31| 0.03|
## [1] "#####"
## [1] "### Analyzing Plant Height (cm) in KIN19 ###"
## [1] "#####"
## [1] "### Plant Height (cm) not taken in KIN19 moving on..."
## [1] "#####"
## [1] "### Analyzing Heading Date (Days) in KIN20 ###"
## [1] "#####"
## [1] "#####"
## [1] "### Summary of fixed effect model for Heading Date (Days) in KIN20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
## data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -291.5323
##
## $nedf
## [1] 211
##
## $sigma
## [1] 1.792016
##
## $varcomp
## component std.error z.ratio bound %ch
## REP 3.249629e-07 NA NA B NA
## units!R 3.211322e+00 0.3126494 10.27132 P 0
##
## $bic
## [1] 593.7684
## attr(,"parameters")

```

```

## [1] 2
##
## $aic
## [1] 587.0647
## attr("parameters")
## [1] 2
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Summary of random effect model for Heading Date (Days) in KIN20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -563.055
##
## $nedf
## [1] 382
##
## $sigma
## [1] 1.786053
##
## $varcomp
##           component std.error   z.ratio bound %ch
## REP          3.228037e-07      NA        NA    B  NA
## GENOTYPE      7.039821e+00  0.9366913  7.515625    P   0
## units!R       3.189984e+00  0.3091001 10.320230    P   0
##
## $bic
## [1] 1143.946
## attr("parameters")
## [1] 3
##
## $aic
## [1] 1132.11
## attr("parameters")
## [1] 3
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Heading Date (Days) in KIN20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |:-|-----:|----:|
## |h2 |       0.69| 0.04|
## [1] "#####"
## [1] "### Analyzing Visual Rating (0-9) in KIN20 ###"
## [1] "#####"

```

```

## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 84.20
##          : Variance heterogeneity factor (Deviance/df): 0.40 assuming 210 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 189.59
##          : Variance heterogeneity factor (Deviance/df): 0.50 assuming 382 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Visual Rating (0-9) in KIN20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##
## $loglik
## [1] 48.24461
##
## $nedf
## [1] 210
##
## $sigma
## [1] 1
##
## $varcomp
##          component std.error z.ratio bound %ch
## REP          0.006077932 0.01023055 0.5940961    P 0.1
## units!R 1.000000000      NA      NA      F 0.0
##
## $bic
## [1] -91.14211
## attr("parameters")
## [1] 1
##
## $aic
## [1] -94.48922
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 84.20462
##
## $heterogeneity
## [1] 0.4009744
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 84.20
##          : Variance heterogeneity factor (Deviance/df): 0.40 assuming 210 df.
## [1] "#####"
## [1] "### Summary of random effect model for Visual Rating (0-9) in KIN20 ###"

```

```

## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       family = asr_poisson(), data = pheno_subset, maxit = 75,
##       trace = FALSE)
##
## $loglik
## [1] 137.4752
##
## $nedf
## [1] 382
##
## $sigma
## [1] 1
##
## $varcomp
##          component  std.error  z.ratio bound %ch
## REP          0.006113495 0.01028347 0.5944974    P    0
## GENOTYPE      0.044629658 0.01444070 3.0905474    P    0
## units!R       1.000000000          NA          NA    F    0
##
## $bic
## [1] -263.0596
## attr("parameters")
## [1] 2
##
## $aic
## [1] -270.9505
## attr("parameters")
## [1] 2
##
## $deviance
## [1] 189.5879
##
## $heterogeneity
## [1] 0.4963034
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Visual Rating (0-9) in KIN20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |---|-----:|----:|
## |h2 |       0.04| 0.01|
## [1] "#####"
## [1] "### Analyzing Fusarium Damaged Kernels (%) in KIN20 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 855.33
##          : Variance heterogeneity factor (Deviance/df): 4.03 assuming 212 df.

```

```

## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 875.53
## : Variance heterogeneity factor (Deviance/df): 2.28 assuming 384 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Fusarium Damaged Kernels (%) in KIN20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
## family = asr_poisson(), data = pheno_subset, maxit = 75,
## trace = FALSE)
##
## $loglik
## [1] -93.81207
##
## $nedf
## [1] 212
##
## $sigma
## [1] 1
##
## $varcomp
## component std.error z.ratio bound %ch
## REP 0.003800331 0.005320064 0.7143393 P 2.9
## units!R 1.000000000 NA NA F 0.0
##
## $bic
## [1] 192.9807
## attr("parameters")
## [1] 1
##
## $aic
## [1] 189.6241
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 855.3259
##
## $heterogeneity
## [1] 4.034556
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 855.32
## : Variance heterogeneity factor (Deviance/df): 4.03 assuming 212 df.
## [1] "#####"
## [1] "### Summary of random effect model for Fusarium Damaged Kernels (%) in KIN20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
## family = asr_poisson(), data = pheno_subset, maxit = 75,

```

```

##      trace = FALSE)
##
## $loglik
## [1] -38.59881
##
## $nedf
## [1] 384
##
## $sigma
## [1] 1
##
## $varcomp
##      component  std.error  z.ratio bound %ch
## REP      0.003743996 0.00547303 0.684081    P    0
## GENOTYPE 0.181155205 0.02142155 8.456682    P    0
## units!R  1.000000000      NA      NA      F    0
##
## $bic
## [1] 89.09891
## attr("parameters")
## [1] 2
##
## $aic
## [1] 81.19763
## attr("parameters")
## [1] 2
##
## $deviance
## [1] 875.5314
##
## $heterogeneity
## [1] 2.28003
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Fusarium Damaged Kernels (%) in KIN20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |---|-----:|----:|
## |h2 |      0.15| 0.02|
## [1] "#####"
## [1] "### Analyzing DON Content (PPM) in KIN20 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 248.45
##          : Variance heterogeneity factor (Deviance/df): 1.49 assuming 167 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 285.49
##          : Variance heterogeneity factor (Deviance/df): 0.84 assuming 339 df.

```

```

## [1] "#####"
## [1] "### Summary of fixed effect model for DON Content (PPM) in KIN20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##       family = asr_poisson(), data = pheno_subset, maxit = 75,
##       trace = FALSE)
##
## $loglik
## [1] 10.62106
##
## $nedf
## [1] 167
##
## $sigma
## [1] 1
##
## $varcomp
##      component  std.error  z.ratio bound  %ch
## REP      0.01998226 0.02337532 0.8548443    P 13.4
## units!R 1.00000000      NA      NA      F 0.0
##
## $bic
## [1] -16.12413
## attr("parameters")
## [1] 1
##
## $aic
## [1] -19.24212
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 248.4495
##
## $heterogeneity
## [1] 1.487722
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 248.44
##          : Variance heterogeneity factor (Deviance/df): 1.49 assuming 167 df.
## [1] "#####"
## [1] "### Summary of random effect model for DON Content (PPM) in KIN20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       family = asr_poisson(), data = pheno_subset, maxit = 75,
##       trace = FALSE)
##
## $loglik
## [1] 35.09985

```

```

##
## $nedf
## [1] 339
##
## $sigma
## [1] 1
##
## $varcomp
##      component  std.error  z.ratio bound %ch
## REP      0.01957144 0.02849703 0.6867889    P    0
## GENOTYPE 0.22546886 0.03060720 7.3665304    P    0
## units!R  1.00000000          NA          NA    F    0
##
## $bic
## [1] -58.5477
## attr("parameters")
## [1] 2
##
## $aic
## [1] -66.1997
## attr("parameters")
## [1] 2
##
## $deviance
## [1] 285.4916
##
## $heterogeneity
## [1] 0.842158
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of DON Content (PPM) in KIN20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## | :--|-----:|----:|
## | h2 |      0.18| 0.02|
## [1] "#####"
## [1] "### Analyzing Plant Height (cm) in KIN20 ###"
## [1] "#####"
## [1] "#####"
## [1] "### Summary of fixed effect model for Plant Height (cm) in KIN20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##      data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -450.4693
##
## $nedf
## [1] 213

```

```

##
## $sigma
## [1] 3.721233
##
## $varcomp
##      component std.error    z.ratio bound %ch
## REP      0.1016584 0.2439487  0.4167204    P 1.1
## units!R 13.8475743 1.3450160 10.2954717    P 0.0
##
## $bic
## [1] 911.6612
## attr(,"parameters")
## [1] 2
##
## $aic
## [1] 904.9386
## attr(,"parameters")
## [1] 2
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Summary of random effect model for Plant Height (cm) in KIN20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##      data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -860.7495
##
## $nedf
## [1] 385
##
## $sigma
## [1] 3.71998
##
## $varcomp
##      component std.error    z.ratio bound %ch
## REP      0.08868932 0.2276998  0.3895011    P 0
## GENOTYPE 35.36943109 4.5876196  7.7097567    P 0
## units!R 13.83825123 1.3431103 10.3031385    P 0
##
## $bic
## [1] 1739.359
## attr(,"parameters")
## [1] 3
##
## $aic
## [1] 1727.499
## attr(,"parameters")
## [1] 3
##
## attr(,"class")

```

```

## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Plant Height (cm) in KIN20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |--|-----:|----:|
## |h2 |       0.72| 0.03|
## [1] "#####"
## [1] "### Analyzing Heading Date (Days) in RAL19 ###"
## [1] "#####"
## [1] "#####"
## [1] "### Summary of fixed effect model for Heading Date (Days) in RAL19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##       data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -226.9481
##
## $nedf
## [1] 172
##
## $sigma
## [1] 1.618647
##
## $varcomp
##      component std.error  z.ratio bound %ch
## REP      0.0652285 0.1142786 0.5707851    P 0.2
## units!R 2.6200181 0.2833498 9.2465841    P 0.0
##
## $bic
## [1] 464.1913
## attr("parameters")
## [1] 2
##
## $aic
## [1] 457.8963
## attr("parameters")
## [1] 2
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Summary of random effect model for Heading Date (Days) in RAL19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -451.3005

```

```

##
## $nedf
## [1] 337
##
## $sigma
## [1] 1.619546
##
## $varcomp
##      component std.error  z.ratio bound %ch
## REP      0.07726578 0.1315088 0.5875332    P    0
## GENOTYPE 4.07182787 0.6106226 6.6683212    P    0
## units!R  2.62293047 0.2834334 9.2541332    P    0
##
## $bic
## [1] 920.0612
## attr(,"parameters")
## [1] 3
##
## $aic
## [1] 908.6009
## attr(,"parameters")
## [1] 3
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Heading Date (Days) in RAL19 ###"
## [1] "#####"
##
##
## | | Estimate| SE|
## |--|-----:|----:|
## |h2 |      0.61| 0.05|
## [1] "#####"
## [1] "### Analyzing Visual Rating (0-9) in RAL19 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 30.85
##          : Variance heterogeneity factor (Deviance/df): 0.18 assuming 169 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 146.15
##          : Variance heterogeneity factor (Deviance/df): 0.44 assuming 334 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Visual Rating (0-9) in RAL19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##
## $loglik
## [1] 31.72525

```

```

##
## $nedf
## [1] 169
##
## $sigma
## [1] 1
##
## $varcomp
##           component std.error z.ratio bound %ch
## REP      1.011929e-07      NA      NA      B  NA
## units!R  1.000000e+00      NA      NA      F   0
##
## $bic
## [1] -58.32059
## attr("parameters")
## [1] 1
##
## $aic
## [1] -61.45049
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 30.85084
##
## $heterogeneity
## [1] 0.1825493
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 30.85
##           : Variance heterogeneity factor (Deviance/df): 0.18 assuming 169 df.
## [1] "#####"
## [1] "### Summary of random effect model for Visual Rating (0-9) in RAL19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       family = asr_poisson(), data = pheno_subset, maxit = 75,
##       trace = FALSE)
##
## $loglik
## [1] 136.7682
##
## $nedf
## [1] 334
##
## $sigma
## [1] 1
##
## $varcomp
##           component std.error z.ratio bound %ch
## REP      1.011929e-07      NA      NA      B  NA

```

```

## GENOTYPE 3.354999e-03 0.01098041 0.3055439      P 1.7
## units!R 1.000000e+00      NA      NA      F 0.0
##
## $bic
## [1] -261.9141
## attr("parameters")
## [1] 2
##
## $aic
## [1] -269.5364
## attr("parameters")
## [1] 2
##
## $deviance
## [1] 146.1499
##
## $heterogeneity
## [1] 0.4375746
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Visual Rating (0-9) in RAL19 ###"
## [1] "#####"
##
##
## | | Estimate| SE|
## |:--|-----:|----:|
## |h2 | | 0| 0.01|
## [1] "#####"
## [1] "### Analyzing Fusarium Damaged Kernels (%) in RAL19 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 1415.37
## : Variance heterogeneity factor (Deviance/df): 8.00 assuming 177 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 1432.18
## : Variance heterogeneity factor (Deviance/df): 4.19 assuming 342 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Fusarium Damaged Kernels (%) in RAL19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
## family = asr_poisson(), data = pheno_subset, maxit = 75,
## trace = FALSE)
##
## $loglik
## [1] -393.6553
##
## $nedf
## [1] 177
##

```

```

## $sigma
## [1] 1
##
## $varcomp
##      component    std.error    z.ratio bound %ch
## REP      0.002464195 0.003662132 0.6728855    P 0.9
## units!R 1.000000000          NA          NA    F 0.0
##
## $bic
## [1] 792.4868
## attr("parameters")
## [1] 1
##
## $aic
## [1] 789.3107
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 1415.368
##
## $heterogeneity
## [1] 7.996429
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 1415.37
##          : Variance heterogeneity factor (Deviance/df): 8.00 assuming 177 df.
## [1] "#####"
## [1] "### Summary of random effect model for Fusarium Damaged Kernels (%) in RAL19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##
## $loglik
## [1] -321.8327
##
## $nedf
## [1] 342
##
## $sigma
## [1] 1
##
## $varcomp
##      component    std.error    z.ratio bound %ch
## REP      0.002453048 0.003693605 0.6641337    P 0
## GENOTYPE 0.165882008 0.020066759 8.2665071    P 0
## units!R 1.000000000          NA          NA    F 0
##
## $bic

```

```

## [1] 655.3351
## attr("parameters")
## [1] 2
##
## $aic
## [1] 647.6654
## attr("parameters")
## [1] 2
##
## $deviance
## [1] 1432.177
##
## $heterogeneity
## [1] 4.187653
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Fusarium Damaged Kernels (%) in RAL19 ###"
## [1] "#####"
##
##
## | | Estimate| SE|
## |--|-----:|----:|
## |h2 | 0.14| 0.01|
## [1] "#####"
## [1] "### Analyzing DON Content (PPM) in RAL19 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 1252.01
## : Variance heterogeneity factor (Deviance/df): 7.07 assuming 177 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 1269.30
## : Variance heterogeneity factor (Deviance/df): 3.71 assuming 342 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for DON Content (PPM) in RAL19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
## family = asr_poisson(), data = pheno_subset, maxit = 75,
## trace = FALSE)
##
## $loglik
## [1] -376.627
##
## $nedf
## [1] 177
##
## $sigma
## [1] 1
##
## $varcomp

```

```

##          component  std.error  z.ratio bound %ch
## REP      0.03711523 0.05281526 0.7027369    P 0.2
## units!R  1.00000000      NA      NA      F 0.0
##
## $bic
## [1] 758.4302
## attr(,"parameters")
## [1] 1
##
## $aic
## [1] 755.254
## attr(,"parameters")
## [1] 1
##
## $deviance
## [1] 1252.009
##
## $heterogeneity
## [1] 7.0735
##
## attr(,"class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 1252.01
##          : Variance heterogeneity factor (Deviance/df): 7.07 assuming 177 df.
## [1] "#####"
## [1] "### Summary of random effect model for DON Content (PPM) in RAL19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##
## $loglik
## [1] -379.1405
##
## $nedf
## [1] 342
##
## $sigma
## [1] 1
##
## $varcomp
##          component  std.error  z.ratio bound %ch
## REP      0.0370263 0.05283733 0.7007602    P 0
## GENOTYPE 0.3792796 0.04606880 8.2328954    P 0
## units!R  1.0000000      NA      NA      F 0
##
## $bic
## [1] 769.9506
## attr(,"parameters")
## [1] 2
##

```

```

## $aic
## [1] 762.2809
## attr(,"parameters")
## [1] 2
##
## $deviance
## [1] 1269.304
##
## $heterogeneity
## [1] 3.711416
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of DON Content (PPM) in RAL19 ###"
## [1] "#####"
##
##
## | | Estimate| SE|
## |--|-----:|----:|
## |h2 | 0.27| 0.02|
## [1] "#####"
## [1] "### Analyzing Plant Height (cm) in RAL19 ###"
## [1] "#####"
## [1] "### Plant Height (cm) not taken in RAL19 moving on..."
## [1] "#####"
## [1] "### Analyzing Heading Date (Days) in RAL20 ###"
## [1] "#####"
## [1] "#####"
## [1] "### Summary of fixed effect model for Heading Date (Days) in RAL20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
## data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -271.9477
##
## $nedf
## [1] 201
##
## $sigma
## [1] 1.713842
##
## $varcomp
## component std.error z.ratio bound %ch
## REP 0.3601125 0.5186706 0.694299 P 1.7
## units!R 2.9372554 0.2937382 9.999570 P 0.0
##
## $bic
## [1] 554.5021
## attr(,"parameters")
## [1] 2
##

```

```

## $aic
## [1] 547.8954
## attr(,"parameters")
## [1] 2
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Summary of random effect model for Heading Date (Days) in RAL20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -477.845
##
## $nedf
## [1] 372
##
## $sigma
## [1] 1.700184
##
## $varcomp
##      component std.error  z.ratio bound %ch
## REP      0.358417 0.5290701  0.677447    P    0
## GENOTYPE  2.598717 0.4490082  5.787683    P    0
## units!R   2.890626 0.2856277 10.120255    P    0
##
## $bic
## [1] 973.4467
## attr(,"parameters")
## [1] 3
##
## $aic
## [1] 961.69
## attr(,"parameters")
## [1] 3
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Heading Date (Days) in RAL20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |:-|-----:|----:|
## |h2 |       0.47| 0.06|
## [1] "#####"
## [1] "### Analyzing Visual Rating (0-9) in RAL20 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models

```

```

## Notice : Deviance from GLM fit: 58.18
##          : Variance heterogeneity factor (Deviance/df): 0.29 assuming 202 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 156.32
##          : Variance heterogeneity factor (Deviance/df): 0.42 assuming 373 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Visual Rating (0-9) in RAL20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##
## $loglik
## [1] 26.53147
##
## $nedf
## [1] 202
##
## $sigma
## [1] 1
##
## $varcomp
##          component std.error z.ratio bound %ch
## REP      1.011929e-07      NA      NA      B  NA
## units!R  1.000000e+00      NA      NA      F   0
##
## $bic
## [1] -47.75467
## attr("parameters")
## [1] 1
##
## $aic
## [1] -51.06294
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 58.1805
##
## $heterogeneity
## [1] 0.2880223
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 58.18
##          : Variance heterogeneity factor (Deviance/df): 0.29 assuming 202 df.
## [1] "#####"
## [1] "### Summary of random effect model for Visual Rating (0-9) in RAL20 ###"
## [1] "#####"
## $call

```

```

## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       family = asr_poisson(), data = pheno_subset, maxit = 75,
##       trace = FALSE)
##
## $loglik
## [1] 112.667
##
## $nedf
## [1] 373
##
## $sigma
## [1] 1
##
## $varcomp
##           component  std.error  z.ratio bound %ch
## REP          1.011929e-07         NA         NA    B  NA
## GENOTYPE      3.649099e-02  0.01428156  2.555113    P 0.2
## units!R       1.000000e+00         NA         NA    F 0.0
##
## $bic
## [1] -213.4908
## attr("parameters")
## [1] 2
##
## $aic
## [1] -221.334
## attr("parameters")
## [1] 2
##
## $deviance
## [1] 156.3216
##
## $heterogeneity
## [1] 0.4190927
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Visual Rating (0-9) in RAL20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |---|-----:|----:|
## |h2 |       0.04| 0.01|
## [1] "#####"
## [1] "### Analyzing Fusarium Damaged Kernels (%) in RAL20 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 897.60
##          : Variance heterogeneity factor (Deviance/df): 4.56 assuming 197 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models

```

```

## Notice : Deviance from GLM fit: 916.99
##          : Variance heterogeneity factor (Deviance/df): 2.49 assuming 368 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Fusarium Damaged Kernels (%) in RAL20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##
## $loglik
## [1] -214.9987
##
## $nedf
## [1] 197
##
## $sigma
## [1] 1
##
## $varcomp
##          component  std.error  z.ratio bound %ch
## REP          0.007070315 0.01035858 0.6825567    P 0.3
## units!R 1.000000000      NA      NA      F 0.0
##
## $bic
## [1] 435.2806
## attr("parameters")
## [1] 1
##
## $aic
## [1] 431.9974
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 897.603
##
## $heterogeneity
## [1] 4.556361
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 897.60
##          : Variance heterogeneity factor (Deviance/df): 4.56 assuming 197 df.
## [1] "#####"
## [1] "### Summary of random effect model for Fusarium Damaged Kernels (%) in RAL20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##

```

```

## $loglik
## [1] -206.7578
##
## $nedf
## [1] 368
##
## $sigma
## [1] 1
##
## $varcomp
##          component  std.error  z.ratio bound %ch
## REP          0.006891546 0.01015117 0.6788916    P    0
## GENOTYPE      0.298449350 0.03594514 8.3029116    P    0
## units!R       1.000000000          NA          NA    F    0
##
## $bic
## [1] 425.3319
## attr(,"parameters")
## [1] 2
##
## $aic
## [1] 417.5157
## attr(,"parameters")
## [1] 2
##
## $deviance
## [1] 916.9876
##
## $heterogeneity
## [1] 2.491814
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Fusarium Damaged Kernels (%) in RAL20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |:-|-----:|----:|
## |h2 |       0.23| 0.02|
## [1] "#####"
## [1] "### Analyzing DON Content (PPM) in RAL20 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 99.30
##          : Variance heterogeneity factor (Deviance/df): 1.07 assuming 93 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 166.43
##          : Variance heterogeneity factor (Deviance/df): 0.65 assuming 257 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for DON Content (PPM) in RAL20 ###"

```

```

## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##       family = asr_poisson(), data = pheno_subset, maxit = 75,
##       trace = FALSE)
##
## $loglik
## [1] -29.81208
##
## $nedf
## [1] 93
##
## $sigma
## [1] 1
##
## $varcomp
##      component  std.error  z.ratio bound %ch
## REP      0.004069305 0.01078283 0.3773876    P    0
## units!R 1.000000000      NA      NA      F    0
##
## $bic
## [1] 64.15675
## attr("parameters")
## [1] 1
##
## $aic
## [1] 61.62415
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 99.29714
##
## $heterogeneity
## [1] 1.067711
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 99.30
##          : Variance heterogeneity factor (Deviance/df): 1.07 assuming 93 df.
## [1] "#####"
## [1] "### Summary of random effect model for DON Content (PPM) in RAL20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       family = asr_poisson(), data = pheno_subset, maxit = 75,
##       trace = FALSE)
##
## $loglik
## [1] -42.69198
##
## $nedf

```

```

## [1] 257
##
## $sigma
## [1] 1
##
## $varcomp
##          component    std.error    z.ratio bound %ch
## REP          0.003484609 0.009130419 0.3816484    P    0
## GENOTYPE      0.258048429 0.049539161 5.2089786    P    0
## units!R       1.000000000          NA          NA    F    0
##
## $bic
## [1] 96.48211
## attr(,"parameters")
## [1] 2
##
## $aic
## [1] 89.38396
## attr(,"parameters")
## [1] 2
##
## $deviance
## [1] 166.4268
##
## $heterogeneity
## [1] 0.6475751
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of DON Content (PPM) in RAL20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |---|-----:|----:|
## |h2 |       0.21| 0.03|
## [1] "#####"
## [1] "### Analyzing Plant Height (cm) in RAL20 ###"
## [1] "#####"
## [1] "#####"
## [1] "### Summary of fixed effect model for Plant Height (cm) in RAL20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##       data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -415.6035
##
## $nedf
## [1] 202
##
## $sigma

```

```

## [1] 3.486223
##
## $varcomp
##      component std.error    z.ratio bound %ch
## REP      0.1319001 0.2775105  0.4752977    P 0.9
## units!R 12.1537527 1.2123660 10.0248218    P 0.0
##
## $bic
## [1] 841.8236
## attr("parameters")
## [1] 2
##
## $aic
## [1] 835.207
## attr("parameters")
## [1] 2
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Summary of random effect model for Plant Height (cm) in RAL20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##      data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -797.0847
##
## $nedf
## [1] 372
##
## $sigma
## [1] 3.483404
##
## $varcomp
##      component std.error    z.ratio bound %ch
## REP      0.1360385 0.2853182  0.4767958    P  0
## GENOTYPE 25.7697226 3.4914451  7.3808185    P  0
## units!R 12.1341043 1.2087180 10.0388215    P  0
##
## $bic
## [1] 1611.926
## attr("parameters")
## [1] 3
##
## $aic
## [1] 1600.169
## attr("parameters")
## [1] 3
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"

```

```

## [1] "### Narrow-sense heritability of Plant Height (cm) in RAL20 ###"
## [1] "#####"
##
##
## | | Estimate| SE|
## |--|-----:|----:|
## |h2 | 0.68| 0.04|
## [1] "#####"
## [1] "### Analyzing Heading Date (Days) in WAR19 ###"
## [1] "#####"
## [1] "### Heading Date (Days) not taken in WAR19 moving on..."
## [1] "#####"
## [1] "### Analyzing Visual Rating (0-9) in WAR19 ###"
## [1] "#####"
## [1] "### Visual Rating (0-9) not taken in WAR19 moving on..."
## [1] "#####"
## [1] "### Analyzing Fusarium Damaged Kernels (%) in WAR19 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 1823.14
## : Variance heterogeneity factor (Deviance/df): 5.38 assuming 339 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Fusarium Damaged Kernels (%) in WAR19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
## G.param = fit1$G.param, R.param = fit1$R.param, data = pheno_subset,
## maxit = 75, trace = FALSE)
##
## $loglik
## [1] -627.1784
##
## $nedf
## [1] 174
##
## $sigma
## [1] 15.95175
##
## $varcomp
## component std.error z.ratio bound %ch
## REP 2.756883 6.037772 0.4566061 P 0
## units!R 254.458308 27.359528 9.3005371 P 0
##
## $bic
## [1] 1264.675
## attr("parameters")
## [1] 2
##
## $aic
## [1] 1258.357
## attr("parameters")

```

```

## [1] 2
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Summary of random effect model for Fusarium Damaged Kernels (%) in WAR19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       G.param = fit2$G.param, R.param = fit2$R.param, data = pheno_subset,
##       maxit = 75, trace = FALSE)
##
## $loglik
## [1] -1213.425
##
## $nedf
## [1] 339
##
## $sigma
## [1] 15.85862
##
## $varcomp
##      component std.error   z.ratio bound %ch
## REP          3.315164  6.793161 0.4880149    P    0
## GENOTYPE 313.352602 49.582143 6.3198681    P    0
## units!R   251.495870 26.804739 9.3825151    P    0
##
## $bic
## [1] 2444.328
## attr("parameters")
## [1] 3
##
## $aic
## [1] 2432.85
## attr("parameters")
## [1] 3
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Fusarium Damaged Kernels (%) in WAR19 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |:-|-----:|----:|
## |h2 |       0.55| 0.05|
## [1] "#####"
## [1] "### Analyzing DON Content (PPM) in WAR19 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 139.80
##       : Variance heterogeneity factor (Deviance/df): 0.80 assuming 175 df.

```

```

## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 173.41
## : Variance heterogeneity factor (Deviance/df): 0.51 assuming 340 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for DON Content (PPM) in WAR19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
## family = asr_poisson(), data = pheno_subset, maxit = 75,
## trace = FALSE)
##
## $loglik
## [1] 40.11346
##
## $nedf
## [1] 175
##
## $sigma
## [1] 1
##
## $varcomp
## component std.error z.ratio bound %ch
## REP 0.02672652 0.03442627 0.776341 P 7.7
## units!R 1.00000000 NA NA F 0.0
##
## $bic
## [1] -75.06213
## attr("parameters")
## [1] 1
##
## $aic
## [1] -78.22691
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 139.7978
##
## $heterogeneity
## [1] 0.7988448
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 139.79
## : Variance heterogeneity factor (Deviance/df): 0.80 assuming 175 df.
## [1] "#####"
## [1] "### Summary of random effect model for DON Content (PPM) in WAR19 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
## family = asr_poisson(), data = pheno_subset, maxit = 75,

```

```

##      trace = FALSE)
##
## $loglik
## [1] 53.35839
##
## $nedf
## [1] 340
##
## $sigma
## [1] 1
##
## $varcomp
##      component  std.error  z.ratio bound %ch
## REP      0.02714219 0.03938714 0.6891132    P    0
## GENOTYPE 0.26265704 0.03613750 7.2682687    P    0
## units!R  1.00000000          NA          NA    F    0
##
## $bic
## [1] -95.0589
## attr("parameters")
## [1] 2
##
## $aic
## [1] -102.7168
## attr("parameters")
## [1] 2
##
## $deviance
## [1] 173.4103
##
## $heterogeneity
## [1] 0.5100304
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of DON Content (PPM) in WAR19 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |:-|-----:|----:|
## |h2 |      0.21| 0.02|
## [1] "#####"
## [1] "### Analyzing Plant Height (cm) in WAR19 ###"
## [1] "#####"
## [1] "### Plant Height (cm) not taken in WAR19 moving on..."
## [1] "#####"
## [1] "### Analyzing Heading Date (Days) in WAR20 ###"
## [1] "#####"
## [1] "#####"
## [1] "### Summary of fixed effect model for Heading Date (Days) in WAR20 ###"
## [1] "#####"
## $call

```

```

## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##       data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -228.4023
##
## $nedf
## [1] 213
##
## $sigma
## [1] 1.313969
##
## $varcomp
##          component  std.error    z.ratio bound %ch
## REP          0.004995412 0.01983856  0.2518031    P 0.3
## units!R      1.726514381 0.16769429 10.2956066    P 0.0
##
## $bic
## [1] 467.5272
## attr(,"parameters")
## [1] 2
##
## $aic
## [1] 460.8046
## attr(,"parameters")
## [1] 2
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Summary of random effect model for Heading Date (Days) in WAR20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       data = pheno_subset, trace = FALSE)
##
## $loglik
## [1] -411.6038
##
## $nedf
## [1] 386
##
## $sigma
## [1] 1.278578
##
## $varcomp
##          component  std.error    z.ratio bound %ch
## REP          0.005884876 0.02037128  0.288881    P 0.1
## GENOTYPE      2.363622532 0.33085283  7.144030    P 0.0
## units!R      1.634760562 0.15383729 10.626556    P 0.0
##
## $bic
## [1] 841.0752
## attr(,"parameters")

```

```

## [1] 3
##
## $aic
## [1] 829.2077
## attr(,"parameters")
## [1] 3
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Heading Date (Days) in WAR20 ###"
## [1] "#####"
##
##
## | | Estimate| SE|
## |--|-----:|----:|
## |h2 | 0.59| 0.04|
## [1] "#####"
## [1] "### Analyzing Visual Rating (0-9) in WAR20 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 181.59
## : Variance heterogeneity factor (Deviance/df): 0.86 assuming 212 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 311.82
## : Variance heterogeneity factor (Deviance/df): 0.81 assuming 385 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Visual Rating (0-9) in WAR20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
## G.param = fit1$G.param, R.param = fit1$R.param, data = pheno_subset,
## maxit = 75, trace = FALSE)
##
## $loglik
## [1] -214.2247
##
## $nedf
## [1] 212
##
## $sigma
## [1] 1.235285
##
## $varcomp
## component std.error z.ratio bound %ch
## REP 0.005279045 0.01883725 0.2802451 P 0
## units!R 1.525929805 0.14856223 10.2713171 P 0
##
## $bic
## [1] 439.1625
## attr(,"parameters")
## [1] 2

```

```

##
## $aic
## [1] 432.4494
## attr(,"parameters")
## [1] 2
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Summary of random effect model for Visual Rating (0-9) in WAR20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       G.param = fit2$G.param, R.param = fit2$R.param, data = pheno_subset,
##       maxit = 75, trace = FALSE)
##
## $loglik
## [1] -371.1921
##
## $nedf
## [1] 385
##
## $sigma
## [1] 1.221756
##
## $varcomp
##          component  std.error   z.ratio bound %ch
## REP          0.006162975 0.01975257  0.3120087    P    0
## GENOTYPE      1.509083191 0.24438335  6.1750655    P    0
## units!R       1.492686523 0.14299102 10.4390230    P    0
##
## $bic
## [1] 760.2439
## attr(,"parameters")
## [1] 3
##
## $aic
## [1] 748.3841
## attr(,"parameters")
## [1] 3
##
## attr(,"class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Visual Rating (0-9) in WAR20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## | :--|-----:|----:|
## | h2 |       0.5| 0.05|
## [1] "#####"
## [1] "### Analyzing Fusarium Damaged Kernels (%) in WAR20 ###"
## [1] "#####"

```

```

## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 741.94
##          : Variance heterogeneity factor (Deviance/df): 3.58 assuming 207 df.
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 761.24
##          : Variance heterogeneity factor (Deviance/df): 2.00 assuming 380 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for Fusarium Damaged Kernels (%) in WAR20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
##        family = asr_poisson(), data = pheno_subset, maxit = 75,
##        trace = FALSE)
##
## $loglik
## [1] -129.2211
##
## $nedf
## [1] 207
##
## $sigma
## [1] 1
##
## $varcomp
##          component      std.error  z.ratio bound %ch
## REP          0.0002927804 0.0007981389 0.3668289    P 0.3
## units!R      1.0000000000          NA          NA    F 0.0
##
## $bic
## [1] 263.7748
## attr("parameters")
## [1] 1
##
## $aic
## [1] 260.4421
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 741.9373
##
## $heterogeneity
## [1] 3.584238
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 741.94
##          : Variance heterogeneity factor (Deviance/df): 3.58 assuming 207 df.
## [1] "#####"
## [1] "### Summary of random effect model for Fusarium Damaged Kernels (%) in WAR20 ###"

```

```

## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
##       family = asr_poisson(), data = pheno_subset, maxit = 75,
##       trace = FALSE)
##
## $loglik
## [1] -118.9353
##
## $nedf
## [1] 380
##
## $sigma
## [1] 1
##
## $varcomp
##          component    std.error    z.ratio bound %ch
## REP          0.0002596108 0.000750982 0.3456951    P    0
## GENOTYPE      0.2956197726 0.035392610 8.3525847    P    0
## units!R       1.0000000000          NA          NA    F    0
##
## $bic
## [1] 249.751
## attr("parameters")
## [1] 2
##
## $aic
## [1] 241.8707
## attr("parameters")
## [1] 2
##
## $deviance
## [1] 761.2369
##
## $heterogeneity
## [1] 2.003255
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of Fusarium Damaged Kernels (%) in WAR20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |---|-----:|----:|
## |h2 |       0.23| 0.02|
## [1] "#####"
## [1] "### Analyzing DON Content (PPM) in WAR20 ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 36.32
##          : Variance heterogeneity factor (Deviance/df): 0.18 assuming 205 df.

```

```

## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 110.25
## : Variance heterogeneity factor (Deviance/df): 0.29 assuming 378 df.
## [1] "#####"
## [1] "### Summary of fixed effect model for DON Content (PPM) in WAR20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ GENOTYPE, random = ~REP, residual = ~units,
## family = asr_poisson(), data = pheno_subset, maxit = 75,
## trace = FALSE)
##
## $loglik
## [1] -17.40842
##
## $nedf
## [1] 205
##
## $sigma
## [1] 1
##
## $varcomp
## component std.error z.ratio bound %ch
## REP 1.011929e-07 NA NA B NA
## units!R 1.000000e+00 NA NA F 0
##
## $bic
## [1] 40.13984
## attr("parameters")
## [1] 1
##
## $aic
## [1] 36.81683
## attr("parameters")
## [1] 1
##
## $deviance
## [1] 36.32193
##
## $heterogeneity
## [1] 0.1771801
##
## attr("class")
## [1] "summary.asreml"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 36.32
## : Variance heterogeneity factor (Deviance/df): 0.18 assuming 205 df.
## [1] "#####"
## [1] "### Summary of random effect model for DON Content (PPM) in WAR20 ###"
## [1] "#####"
## $call
## asreml(fixed = Y ~ 1, random = ~GENOTYPE + REP, residual = ~units,
## family = asr_poisson(), data = pheno_subset, maxit = 75,

```

```

##      trace = FALSE)
##
## $loglik
## [1] 12.84039
##
## $nedf
## [1] 378
##
## $sigma
## [1] 1
##
## $varcomp
##           component  std.error  z.ratio bound %ch
## REP          1.011929e-07         NA         NA    B   NA
## GENOTYPE      1.245587e-01  0.03176933  3.920721    P    0
## units!R       1.000000e+00         NA         NA    F    0
##
## $bic
## [1] -13.811
## attr("parameters")
## [1] 2
##
## $aic
## [1] -21.68079
## attr("parameters")
## [1] 2
##
## $deviance
## [1] 110.2472
##
## $heterogeneity
## [1] 0.2916592
##
## attr("class")
## [1] "summary.asreml"
## [1] "#####"
## [1] "### Narrow-sense heritability of DON Content (PPM) in WAR20 ###"
## [1] "#####"
##
##
## |   | Estimate|   SE|
## |:-|-----:|----:|
## |h2 |       0.11| 0.03|
## [1] "#####"
## [1] "### Analyzing Plant Height (cm) in WAR20 ###"
## [1] "#####"
## [1] "### Plant Height (cm) not taken in WAR20 moving on..."

```

Visualization

Here I visualize the results of the above loops. Output from these loops are pairs plots which show scatter plots fit with a linear regression in the lower triangle. The diagonal of these plots shows the distribution of each trait which is listed as plain text above the distribution with the corresponding name of the environment (KIN = Kinston, NC; RAL = Raleigh, NC; WAR = Warsaw, VA; 19 = 2019; 20 = 2020).

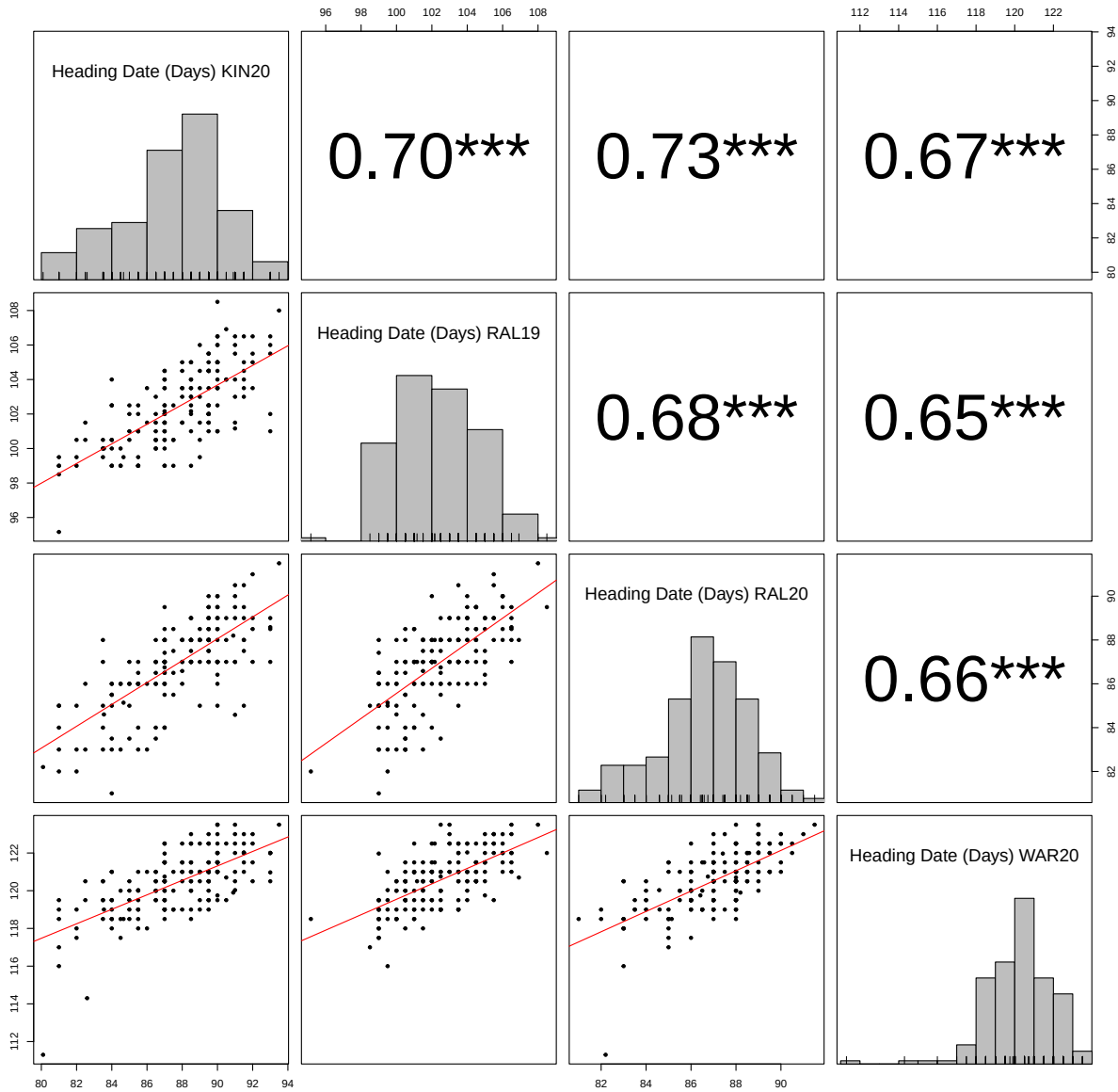
```

# For i in environments and traits
for(i in traits) {
  # Parse blues and h2
  i <- gsub("\\([^\\)]*\\)", "", i)
  i <- trimws(i)
  temp1 <- BLUES_loc[,grep(i,colnames(BLUES_loc))]
  temp2 <- h2_envs[grep(i, h2_envs$Trait),]
  colnames(temp1) <- gsub("_", " ",colnames(temp1))

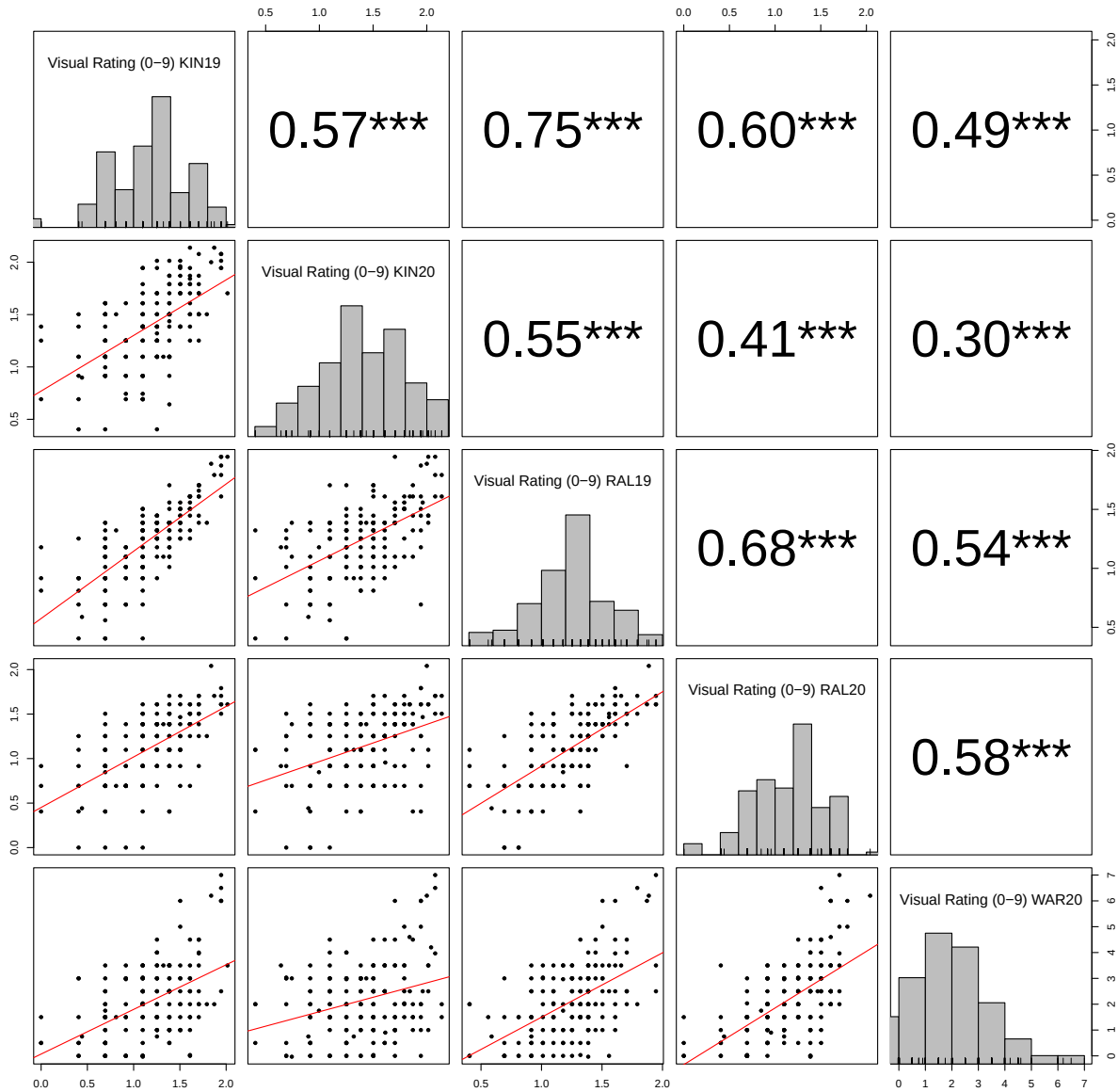
  pairs.panels(temp1,
    hist.col = "gray",
    lm = TRUE,
    stars = TRUE,
    digits = 2,
    density = FALSE,
    ellipses = FALSE,
    main = paste("Visualization of",
                  i,
                  "correlations across environments"))
}

```

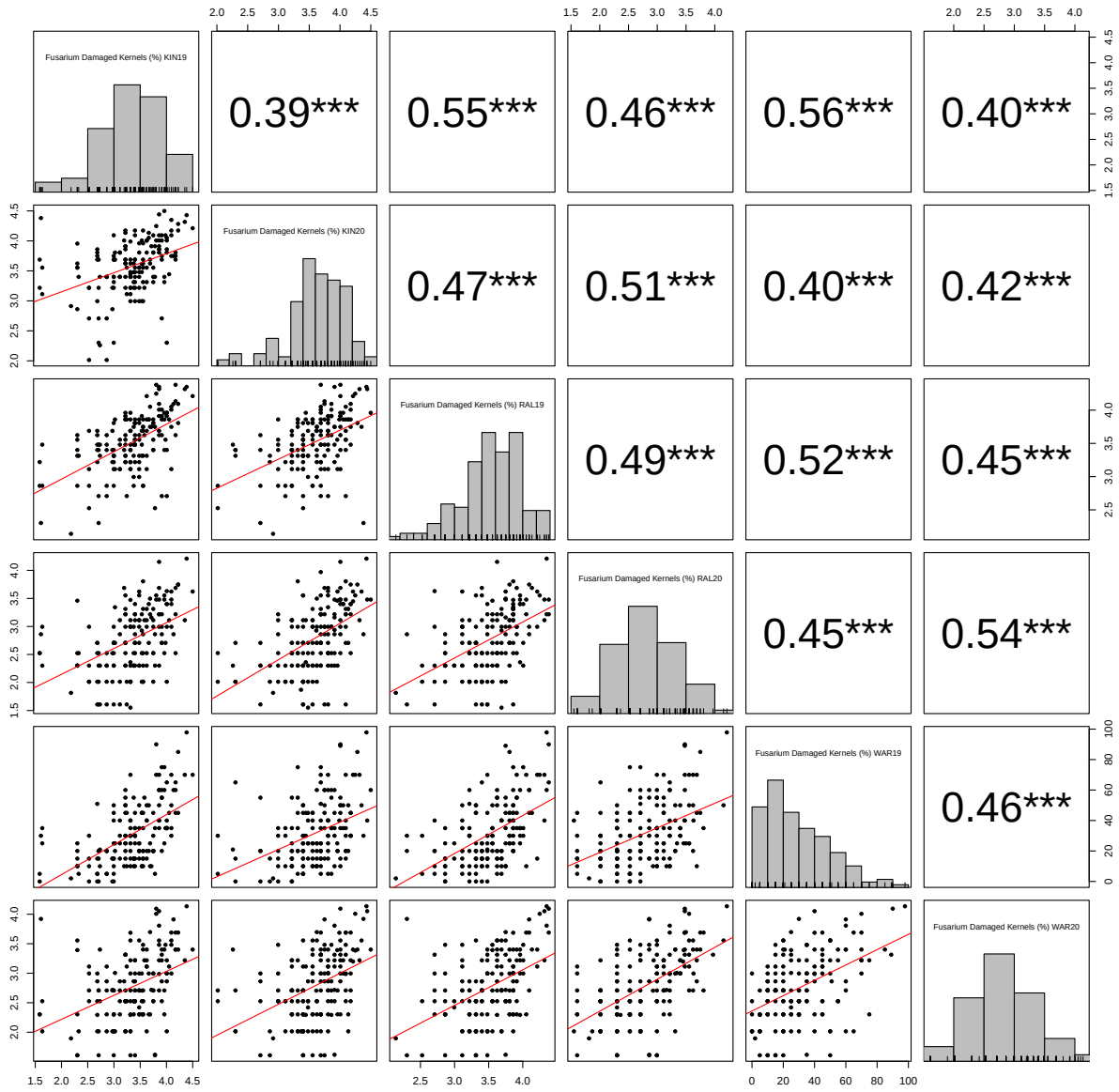
Visualization of Heading Date correlations across environments



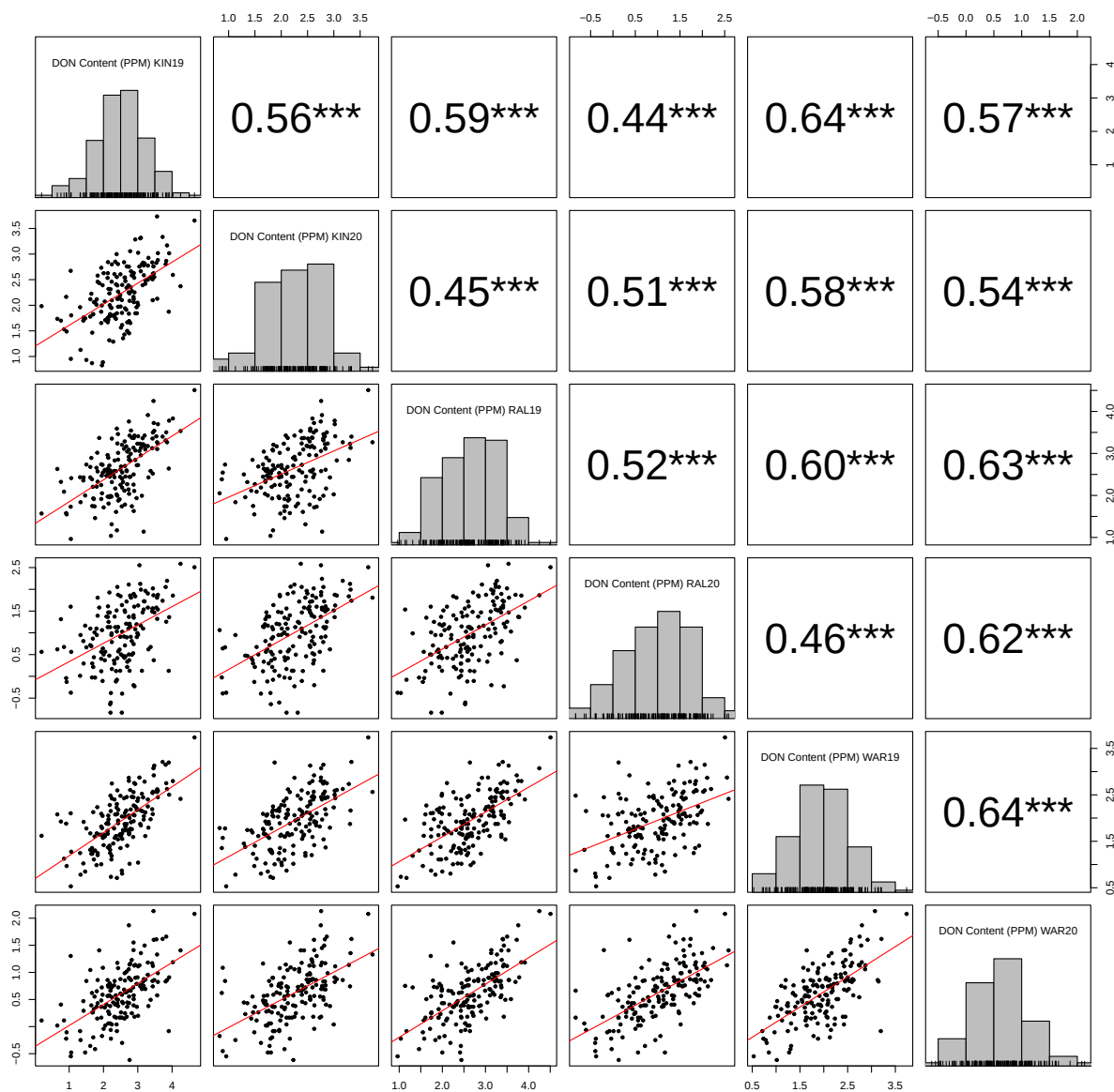
Visualization of Visual Rating correlations across environments

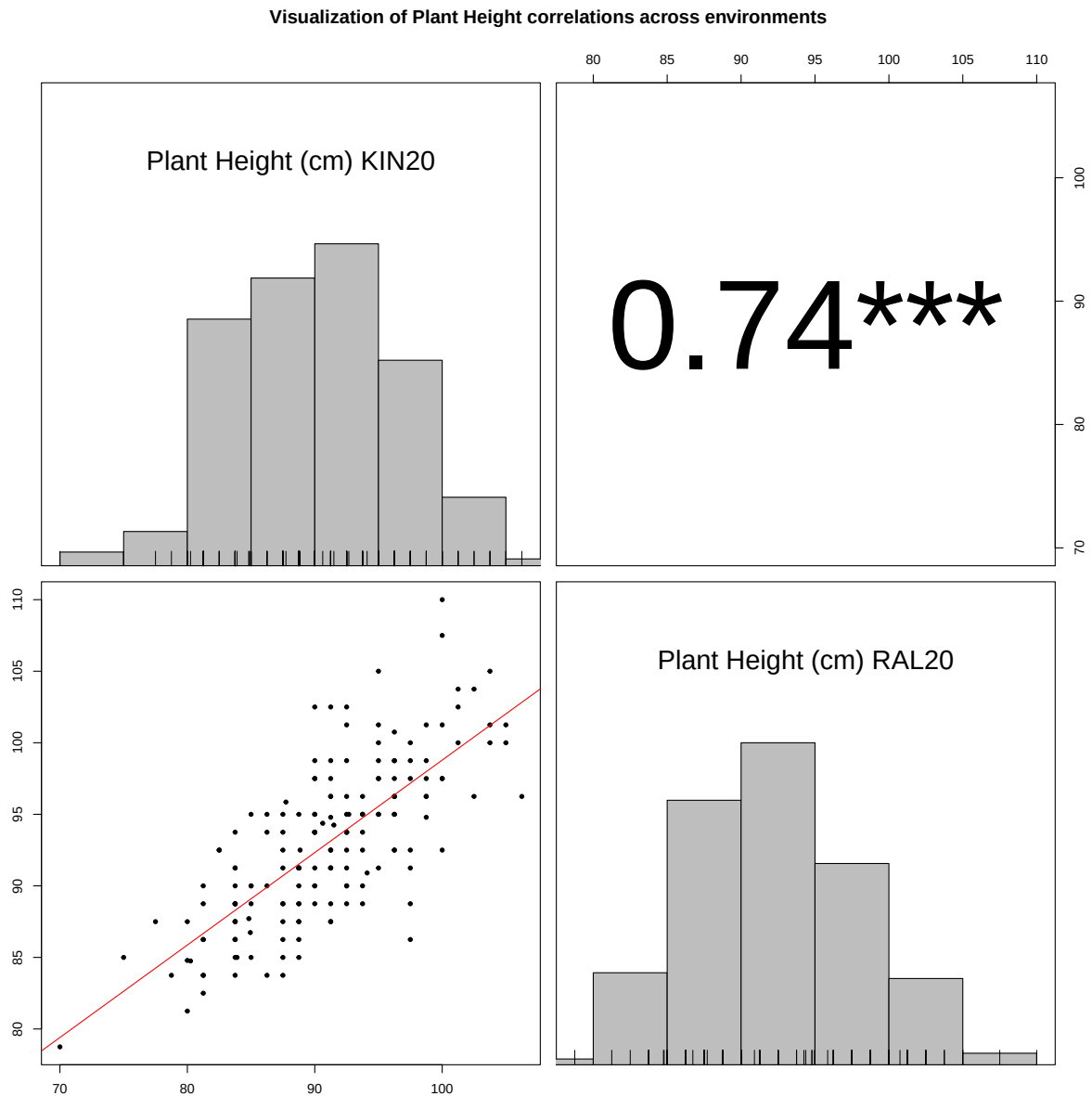


Visualization of Fusarium Damaged Kernels correlations across environments



Visualization of DON Content correlations across environments





Assess Genotype-by-Environment Analysis

Here I run a model for each trait:

$$y_{ijk} = \mu + G_i + E_j + GE_{ij} + er_{jk} + \varepsilon_{ijk}$$

Where G is the fixed genotype effect, E is the fixed environment effect, GE is the fixed genotype by environment interaction, er is the random environment by replication interaction, and ε is the residual error. All random effects are individually and identically distributed.

```
# Define traits
traits <- colnames(pheno)[8:12]
```

```

# Create dataframe for predictions
BLUES_mem <- data.frame(GENOTYPE = unique(pheno$GENOTYPE))

# Run MEMLM loop for each trait
for (i in traits) {

  # Print message for analyzing each trait
  print_message <- paste("### Analyzing", i, "###")
  print(paste(rep("#", nchar(print_message)), collapse = ""))
  print(print_message)
  print(paste(rep("#", nchar(print_message)), collapse = ""))

  # Subset data
  pheno_subset <- na.omit(pheno[, c("GENOTYPE", "YEAR", "ENV", "REP", i)])

  # Redefine data types
  pheno_subset[, 1:4] <- lapply(pheno_subset[, 1:4], as.factor)
  pheno_subset[, 5] <- as.numeric(pheno_subset[, 5])
  colnames(pheno_subset)[5] <- "Y"

  # Run mixed effects model for traits
  if (i %in% c("Visual Rating (0-9)",
              "Fusarium Damaged Kernels (%)",
              "DON Content (PPM)")) {
    # Suppress output from the first model with Poisson distribution
    suppressWarnings(suppressMessages({
      capture.output({
        fit <- asreml(fixed = Y ~ GENOTYPE,
                     random = ~ ENV + GENOTYPE:ENV + REP:ENV,
                     residual = ~ units,
                     data = pheno_subset,
                     family = asr_poisson(),
                     maxit = 75,
                     trace = FALSE)
      }, file = NULL)
    })))

    # Suppress output from the second model with Poisson distribution
    suppressWarnings(suppressMessages({
      capture.output({
        fit2 <- asreml(fixed = Y ~ GENOTYPE + ENV + GENOTYPE:ENV,
                      random = ~ REP:ENV,
                      residual = ~ units,
                      data = pheno_subset,
                      family = asr_poisson(),
                      maxit = 75,
                      trace = FALSE)
      }, file = NULL)
    })))

    # If model fails to converge, fit normal distribution model and suppress output
    if (fit$converge==FALSE | fit2$converge==FALSE){
      suppressWarnings(suppressMessages({

```

```

capture.output({
  fit <- asreml(fixed = Y ~ GENOTYPE,
               random = ~ ENV + GENOTYPE:ENV + REP:ENV,
               residual = ~ units,
               data = pheno_subset,
               maxit = 75,
               trace = FALSE)
}, file = NULL)
}))

suppressWarnings(suppressMessages({
  capture.output({
    fit2 <- asreml(fixed = Y ~ GENOTYPE + ENV + GENOTYPE:ENV ,
                  random = ~ REP:ENV,
                  residual = ~ units,
                  data = pheno_subset,
                  maxit = 75,
                  trace = FALSE)
  }, file = NULL)
}))
}
} else {
  # Suppress output from the model with normal distribution
  suppressWarnings(suppressMessages({
    capture.output({
      fit <- asreml(fixed = Y ~ GENOTYPE,
                   random = ~ ENV + GENOTYPE:ENV + REP:ENV,
                   residual = ~ units,
                   data = pheno_subset,
                   maxit = 75,
                   trace = FALSE)
    }, file = NULL)
  }))

  # Suppress output from the second model with normal distribution
  suppressWarnings(suppressMessages({
    capture.output({
      fit2 <- asreml(fixed = Y ~ GENOTYPE + ENV + GENOTYPE:ENV ,
                    random = ~ REP:ENV,
                    residual = ~ units,
                    data = pheno_subset,
                    maxit = 75,
                    trace = FALSE)
    }, file = NULL)
  }))
}

# Print summary of Wald test for fit2
temp1 <- capture.output({
  wald_output <- as.data.frame(wald(fit2))
}, file = NULL)

print(wald_output)

```

```

# Get predictions for GENOTYPE
pred <- predict(fit, classify = "GENOTYPE")$pvals[, 1:2]
colnames(pred)[2] <- paste(i, "ME", sep = "_")

# Left join predictions to the main dataframe
BLUES_mem <- left_join(BLUES_mem, pred, by = "GENOTYPE")

# Remove temporary objects
rm(pheno_subset, pred)
}

```

```

## [1] "#####"
## [1] "### Analyzing Heading Date (Days) ###"
## [1] "#####"
##           Df      Sum of Sq Wald statistic   Pr(Chisq)
## (Intercept)    1 1.726132e+06    658923.8636 0.000000e+00
## GENOTYPE       175 7.108824e+03     2713.6821 0.000000e+00
## ENV            3 3.245757e+04    12390.1682 0.000000e+00
## GENOTYPE:ENV   505 1.988523e+03      759.0876 1.608491e-12
## residual (MS)  NA 2.619623e+00          NA      NA
## [1] "#####"
## [1] "### Analyzing Visual Rating (0-9) ###"
## [1] "#####"
##           Df Sum of Sq Wald statistic Pr(Chisq)
## (Intercept)    1 6839.4700     6839.4700      0
## GENOTYPE       174 627.3519     627.3519      0
## ENV            4 116.2983     116.2983      0
## GENOTYPE:ENV   671 331.9663     331.9663      1
## residual (MS)  NA 1.0000          NA      NA
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 824.60
##           : Variance heterogeneity factor (Deviance/df): 0.52 assuming 1599 df.
## [1] "#####"
## [1] "### Analyzing Fusarium Damaged Kernels (%) ###"
## [1] "#####"
##           Df      Sum of Sq Wald statistic   Pr(Chisq)
## (Intercept)    1 553457.9837    3376.6121 0.000000e+00
## GENOTYPE       174 387963.5243    2366.9409 0.000000e+00
## ENV            5 42278.8276     257.9405 0.000000e+00
## GENOTYPE:ENV   836 197346.9733    1204.0014 8.881784e-16
## residual (MS)  NA 163.9093          NA      NA
## [1] "#####"
## [1] "### Analyzing DON Content (PPM) ###"
## [1] "#####"
##           Df Sum of Sq Wald statistic Pr(Chisq)
## (Intercept)    1 3542.1302     3542.1302      0
## GENOTYPE       174 8091.7847     8091.7847      0
## ENV            5 390.7614      390.7614      0
## GENOTYPE:ENV   829 1954.9657     1954.9657      0
## residual (MS)  NA 1.0000          NA      NA
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models

```

```
## Notice : Deviance from GLM fit: 2681.79
## : Variance heterogeneity factor (Deviance/df): 1.50 assuming 1790 df.
## [1] "#####"
## [1] "### Analyzing Plant Height (cm) ###"
## [1] "#####"
##           Df      Sum of Sq Wald statistic    Pr(Chisq)
## (Intercept)    1 2.367892e+06   181824.92053 0.000000e+00
## GENOTYPE       172 2.458192e+04    1887.58815 0.000000e+00
## ENV            1 3.631341e+02     27.88422 1.287957e-07
## GENOTYPE:ENV   170 3.475321e+03    266.86178 2.956191e-06
## residual (MS)  NA 1.302292e+01             NA             NA
```

Heritability Variance Component Estimates

Here we will use a fully random version of the above model to assess the per-plot and entry-mean heritability of each trait across environments. Variance components will be shown for each loop.

```
# Make an empty data frame for the heritability estimates
h2s <- data.frame(Trait = character(),
                  Type = character(),
                  Estimation = numeric(),
                  SE = numeric())

# Calculate the harmonic means for each trait
e_traits <- list()
r_traits <- list()

for (j in traits) {
  # Subset data and only take complete cases
  a <- pheno %>%
    drop_na(all_of(j))

  # Get the number of observations per genotype
  obs_per_ind <- aggregate(a[, j] ~ GENOTYPE,
                           data = a,
                           length)

  # Calculate the harmonic mean of replications
  r <- (1 / mean(1 / obs_per_ind$a[, j]))

  # Store harmonic mean in r_traits list
  r_traits[j] <- r

  # Create a new object called e
  e <- c()

  # For each genotype in the phenotypic dataset
  for (i in unique(pheno$GENOTYPE)) {
    # Get all observations for each genotype
    a <- na.omit(pheno[pheno$GENOTYPE == i, c("ENV", j)])

    # Get the number of environments the genotype was observed in
    a <- length(unique(a$ENV))
```

```

    # Calculate harmonic mean for environments
    a <- ifelse(a == 0, 0, 1 / a)

    # Add this value to e
    e <- rbind(e, a)
  }

  # Store harmonic mean of environments in e_traits list
  e_traits[j] <- (1 / mean(e))

  # Clean up temporary variables
  remove(a, obs_per_ind, r, e)
}

# Loop through each trait for heritability estimation
for (i in traits) {
  # Print message for analyzing each trait
  print_message <- paste("### Analyzing", i, "###")
  print(paste(rep("#", nchar(print_message)), collapse = ""))
  print(print_message)
  print(paste(rep("#", nchar(print_message)), collapse = ""))

  # Determine number of locations
  n_loc <- pheno %>%
    select(ENV, YEAR, LOC, all_of(i)) %>%
    drop_na() %>%
    distinct(ENV)
  n_loc <- as.numeric(nrow(n_loc))

  # Determine the number of replications
  n_rep <- n_loc * 2

  # Determine the number of years
  n_year <- pheno %>%
    select(ENV, YEAR, LOC, all_of(i)) %>%
    drop_na() %>%
    distinct(YEAR)
  n_year <- as.numeric(count(n_year))

  # Get the harmonic mean of environments
  e <- as.numeric(e_traits[i])

  # Get the harmonic mean of replications
  r <- as.numeric(r_traits[i])

  # If the trait has been observed over many years
  if (n_year >= 2) {
    # If the trait is a disease reaction trait
    if (i %in% c("Visual Rating (0-9)",
                 "Fusarium Damaged Kernels (%)",
                 "DON Content (PPM)")) {
      # Run model for Poisson distributed traits
      mlm <- asreml(fixed = pheno[, i] ~ 1,

```

```

        random = ~ GENOTYPE +
            ENV +
            GENOTYPE:ENV +
            REP:ENV,
        residual = ~ idv(units),
        data = pheno,
        maxit = 75,
        family = asr_poisson(),
        trace = FALSE)
    } else {
        # Run model for normally distributed traits
        mlm <- asreml(fixed = pheno[, i] ~ 1,
            random = ~ (GENOTYPE) +
                ENV +
                GENOTYPE:ENV+
                REP:ENV,
            residual = ~ idv(units),
            data = pheno,
            maxit = 75,
            trace = FALSE)
    }
    # If the trait has been only observed in one year
} else {
    # Run model for single year data
    mlm <- asreml(fixed = pheno[, i] ~ 1,
        random = ~ (GENOTYPE) +
            ENV +
            GENOTYPE:ENV+
            REP:ENV,
        residual = ~ idv(units),
        data = pheno,
        maxit = 75,
        trace = FALSE)
}

# Print summary of variance components
print(summary(mlm)$varcomp)

# Predict heritability estimates
pph2 <- vpredict(mlm,
    h2 ~ V2 / (V2 + V3 + V4))

emh2 <- vpredict(mlm,
    h2 ~ V2 / (V2 + (V3 / e) + (V4 / (e * r))))

# Bind predictions to h2s data frame
a <- rbind(pph2, emh2)
a <- cbind(data.frame(Trait = c(i, i)),
    data.frame(Type = c("Per-Plot", "Entry-Mean")),
    a)
h2s <- rbind(h2s, a)

# Clean up temporary variables

```

```

remove(mlm, pph2, emh2, n_loc, n_rep, n_year, e, r, a)
}

## [1] "#####"
## [1] "### Analyzing Heading Date (Days) ###"
## [1] "#####"

## Warning in asreml(fixed = pheno[, i] ~ 1, random = ~(GENOTYPE) + ENV + GENOTYPE:ENV + : Warning : Log

##           component      std.error    z.ratio bound %ch
## ENV          247.4376008 201.26565279  1.229408    P 0.3
## REP:ENV        0.1070290  0.08555271  1.251029    P 0.0
## GENOTYPE        3.4600146  0.42313592  8.177076    P 0.0
## GENOTYPE:ENV    0.6396008  0.12336087  5.184795    P 0.0
## units!units    2.5485560  0.12388534 20.571893    P 0.0
## units!R         1.0000000          NA      NA      F 0.0
## [1] "#####"
## [1] "### Analyzing Visual Rating (0-9) ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 644.55
##           : Variance heterogeneity factor (Deviance/df): 0.36 assuming 1773 df.
##           component      std.error    z.ratio bound %ch
## ENV          0.070895622 0.051071336  1.388169    P  0
## REP:ENV        0.001574212 0.001365572  1.152786    P  0
## GENOTYPE        0.090576036 0.011634331  7.785238    P  0
## GENOTYPE:ENV    0.023543544 0.005046725  4.665114    P  0
## units!units    0.383047867 0.017494185 21.895725    P  0
## units!R         1.000000000          NA      NA      F  0
## [1] "#####"
## [1] "### Analyzing Fusarium Damaged Kernels (%) ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 10117.66
##           : Variance heterogeneity factor (Deviance/df): 4.77 assuming 2121 df.
##           component      std.error    z.ratio bound %ch
## ENV          0.108653656 0.070001809  1.552155    P  0
## REP:ENV        0.002328852 0.001934286  1.203985    P  0
## GENOTYPE        0.136799638 0.017279493  7.916878    P  0
## GENOTYPE:ENV    0.037886264 0.007015852  5.400094    P  0
## units!units    5.258609316 0.215433865 24.409390    P  0
## units!R         1.000000000          NA      NA      F  0
## [1] "#####"
## [1] "### Analyzing DON Content (PPM) ###"
## [1] "#####"
## Notice : Distribution and link: Poisson; Log Mu=exp(XB) V=Mu
## Notice : The LogL value is unsuitable for comparing GLM models
## Notice : Deviance from GLM fit: 3112.17
##           : Variance heterogeneity factor (Deviance/df): 1.58 assuming 1964 df.
##           component      std.error    z.ratio bound %ch
## ENV          0.69706385 0.44681455  1.560074    P 0.1
## REP:ENV        0.01674546 0.01052415  1.591146    P 0.0
## GENOTYPE        0.21927507 0.02732240  8.025469    P 0.0

```

```

## GENOTYPE:ENV 0.05815537 0.00825453 7.045268 P 0.1
## units!units 1.82615058 0.07303225 25.004713 P 0.0
## units!R 1.00000000 NA NA F 0.0
## [1] "#####"
## [1] "### Analyzing Plant Height (cm) ###"
## [1] "#####"
##
## component std.error z.ratio bound %ch
## ENV 2.1748182 3.2370663 0.6718485 P 0
## REP:ENV 0.1125161 0.1813375 0.6204792 P 0
## GENOTYPE 27.2939563 3.5201815 7.7535652 P 0
## GENOTYPE:ENV 3.4907961 1.1152603 3.1300282 P 0
## units!units 12.9632033 0.8947734 14.4876946 P 0
## units!R 1.0000000 NA NA F 0

```